

Only TLS Client Connections SHALL be allowed using the clusters and commands in this chapter.

14.3. Description

TLS compatibility will be enabled by the following features.

14.3.1. TLS Certificate Management

This cluster allows for the provisioning of X.509 Root Certificates and the creation of and management of X.509 Client Certificates as used with the TLS protocol implementation.

Nodes wishing to use TLS as part of Matter SHALL support the [TLS Certificate Management Cluster](#).

Since standard Web PKI and TLS policy performs time and date validation of all X.509 certificates in the chain, Nodes supporting TLS Certificate Management SHALL support the [Time Synchronization](#) cluster with either the [NTPClient](#) or [TimeSyncClient](#) feature enabled. See [Section 11.17.11, “Time Synchronization at Commissioning”](#) and [Section 11.17.12, “Time Synchronization during operation”](#) for additional details on how time information gets set and updated.

14.3.1.1. TLS Root CA Certificate Management

Authentication of a server ensures that the client is connecting to the correct server and not to an impostor. TLS uses Root CA Certificates to authenticate a server during the handshake process.

Nodes wishing to provide Root CA Certificate Management SHALL support the [TLS Certificate Management Cluster](#). In addition, Nodes SHALL provide the following.

Certificate Authority ID (CAID)

A Certificate Authority ID ([CAID](#)) is used to uniquely identify a provisioned [TLSRCAC](#) associated with a Fabric on a Node.

Nodes SHALL provide a mechanism where the CAID can be used to get, update, or remove the [TLSRCAC](#) associated with the CAID (see [TLS Certificate Management Commands](#)).

14.3.1.2. TLS Client Certificate Management

Some TLS Servers MAY choose to implement Client Certificate Authentication. To facilitate this, Nodes MAY choose to provision Client Certificate Details ([TLSClientCertificateDetailStruct](#)) which are associated with a Fabric for use in TLS communication. Client Certificate Details (CCD) is used to refer to the combination of client certificate, private key and chain of intermediate certificates that is needed to complete the TLS handshake during Client Authentication.

Nodes wishing to provide Client Certificate Management SHALL support the [TLS Certificate Management Cluster](#).

Client Certificate Details ID (CCDID)

A Client Certificate Details ID ([CCDID](#)) is used to uniquely identify Client Certificate Details associated with a Fabric on a Node.